

Algebra 1
Unit 12
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

For questions 1-3, identify whether the statement is true or false. If it is false, rewrite the statement to make it true.

1. Stem-and-leaf plots are useful to show large quantities of data. *True or False*
Stem-and-leaf plots are useful to show small quantities of data
2. Line plots can show the range of small quantities of data. *True or False*
3. Box plots show overall shape for large quantities of data. *True or False*
4. A survey of 20 high school students was conducted to determine average time in minutes to travel to school. Which graphical displays would best display the data?
Select all that apply.

- ☒ Box plot
- ☒ Line plot
- ☐ Histogram
- ☒ Stem-and-leaf plot

5. Explain your reasoning for question 4.
Answers will vary. Line and Stem-and-Leaf plots are great for displaying individual data values. All three are also great for showing the distribution and the shape for a small quantity of data.
6. The data of the 20 high school students are below.

{20, 8, 12, 31, 9, 9, 11, 4, 23, 17, 16, 10, 10, 7, 21, 19, 7, 18, 15, 10}

Explain which display would be better for the data, a stem-and-leaf plot or a histogram.
Answers may vary. A Stem-and-Leaf plot because the data can be grouped in 10's, individual values can be shown, as well as the shape of the distribution of the dataset.

Appliances-R-Us was analyzing their inventory before their next sale. The frequency table shows which appliances are in stock.

Frequency Table

Type of Appliance	Frequency
Microwave	22
Dishwasher	18
Oven	27
Refrigerator	13

Relative Frequency Table

Type of Appliance	Frequency
Rap	27.5%
Hip hop	22.5%
Rock	33.75%
Other	16.25%

7. In order to create a bar graph, the data from the ☐ relative frequency
☒ frequency table is needed.
8. In order to create a circle graph, the data from the ☒ relative frequency
☐ frequency table is needed.
9. Which of the follow are types of displays for univariate categorical data? *Select all that apply.*

- ☒ Line Plot
 - ☒ Circle Graph
 - ☐ Stem-and-leaf plot
 - ☒ Frequency Table
 - ☐ Box plot
 - ☒ Bar Graph
 - ☐ Histogram

10. Which of the follow are types of displays for univariate numerical data? *Select all that apply.*

- ☒ Line Plot
 - ☐ Circle Graph
 - ☒ Stem-and-leaf plot
 - ☐ Frequency Table
 - ☒ Box plot
 - ☐ Bar Graph
 - ☒ Histogram